Tenant-Customer-Site-Device Model

Tenant-Customer-Site-Device Model

The GCXONE platform uses a 5-level hierarchical structure to organize security assets, manage access control, and ensure data isolation. Understanding this hierarchy is essential for effective platform administration and configuration.

The 5-Level Hierarchy Structure

The hierarchy flows from top to bottom as follows: Tenant → Customer → Site → Device → Camera/Sensor. Each level is nested inside the one above it, creating a clean organizational structure that enforces data isolation at every layer.

Level 1: Tenant

The Tenant is the top-level organization with completely isolated data and configuration.

Complete Data Isolation: Each tenant operates in its own dedicated space.

Independent Configuration: Custom settings, branding, and integrations per tenant.

Subdomain Access: Each tenant gets a unique subdomain such as company.nxgen.cloud.

Multi-Customer Support: A single tenant can serve multiple customers, making it ideal for the MSSP model.

Common use cases include Managed Security Service Providers (MSSPs), large enterprises with multiple divisions, and service providers offering white-label solutions.

Level 2: Customer

The Customer represents individual clients or sub-organizations within a tenant.

Client Organization: A distinct organization or business unit under the tenant.

Isolated Data: Customer data is completely separate from other customers.

Custom Configuration: Site-specific settings and device configurations per customer.

User Management: Customer-level user accounts and permissions.

Common use cases include individual businesses served by an MSSP, different divisions within a large enterprise, and separate legal entities under a parent organization.

Level 3: Site

A Site is a physical location such as a building, facility, or geographic area.

Physical Location: Includes a real-world address and geographic coordinates.

Talos Synchronization: Sites automatically sync with Talos CMS via MQTT messaging. Site names, addresses, contact info, coordinates, and configuration parameters are kept consistent in real time with bidirectional synchronization and error handling.

Device Grouping: All devices at a location are grouped under their site.

Address Mapping: Used for mapping, routing, and location services.

Note: To delete a site, you must be at the Customer level.

Level 4: Device

A Device is the security hardware at a site — such as an NVR, DVR, alarm panel, or IoT gateway.

Device types supported: NVR (IP camera recording), DVR (analog camera recording), Alarm Panels (intrusion detection), and IoT Gateways (smart sensors).

Each device requires the following configuration: a unique Server Unit ID, IP address and port, authentication credentials, and NTP configuration.

Important: Each device must have a unique Server Unit ID. Duplicate IDs cause alarm attribution conflicts.

Level 5: Camera / Sensor

The Camera or Sensor is the individual channel connected to a device.

Sensor types include video cameras, motion sensors, door sensors, and IoT sensors.

Individual Channel: Each sensor occupies a separate channel on its parent device.

Automatic Discovery: Sensors are discovered automatically when a device is added.

Alarm Attribution: Each sensor can independently trigger alarms and events.

Capabilities per sensor may include live stream, playback, PTZ control, and analytics such as motion detection and line crossing.

Common Hierarchy Patterns

Single Organization: One tenant contains one customer, which contains one or more sites.

Multi-Location Business: One tenant contains one customer with multiple sites — ideal for retail chains or enterprises with branch offices.

MSSP Model: One tenant contains multiple customers, each with their own sites and devices — the standard model for managed security service providers.

Integration with Talos CMS

GCXONE and Talos are automatically synchronized by default. Sites created in GCXONE appear directly in Talos. Alarms are attributed to the correct site and device, and events are correlated based on hierarchy relationships.

If a specific site is not fully synced, you can manually force synchronization from the site actions menu using Push to Talos (sends data from GCXONE to Talos) or Pull from Talos (retrieves data from Talos to GCXONE).

Best Practices

Use clear, consistent naming conventions across all levels.

Ensure site addresses are accurate for Talos synchronization.

Always assign unique Server Unit IDs to avoid alarm attribution conflicts.

Organize sites and devices logically to reflect real-world structure.

Document tenant-specific configurations for future reference.

Avoid reusing Server Unit IDs, using vague names, skipping address validation, or ignoring hierarchy when setting permissions.